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0178/03

October/November 2017

2 hours

Candidates answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 100.

This document consists of **18** printed pages and **2** blank pages.



1 A straight line N joins points $E(-1, d)$, $F(1, 1)$ and $G(3, -5)$.

(a) Given that F is the midpoint of EG , find the value of d .

Answer (a) $d = \dots\dots\dots$ [2]

(b) A straight line has gradient 2 and passes through $(0, 5)$.

Write the equation of this line.

Answer (b) $\dots\dots\dots$ [2]

(c) A point H has coordinates $(x, 9)$.

Given that the length of the line joining F and H is 10 units.

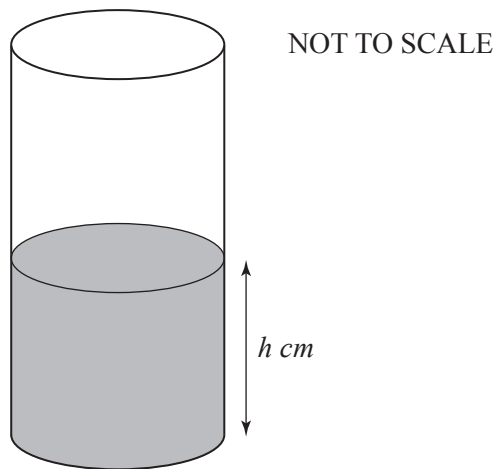
Find two possible values of x .

Answer (c) $x = \dots\dots\dots x = \dots\dots\dots$ [3]

- 2 The diagram shows a cylindrical container of radius 50 cm.

The container is filled with $942\,600\text{ cm}^3$ of water.

The depth of the water is h cm



Use $\pi = 3.142$

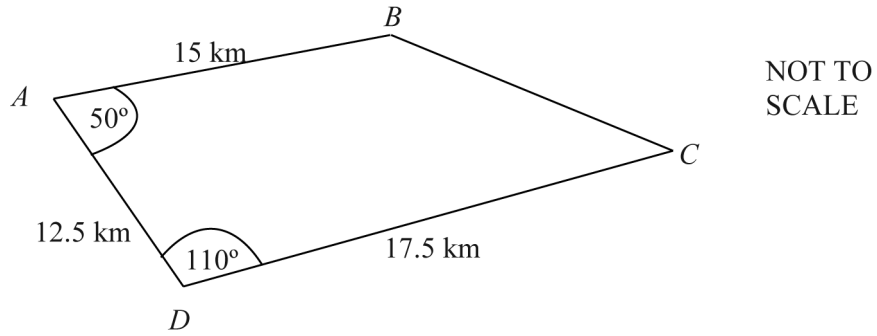
- (a) Show that h is 120 cm.

[2]

- (b) Calculate the area of the curved part of the cylinder that is in contact with water.

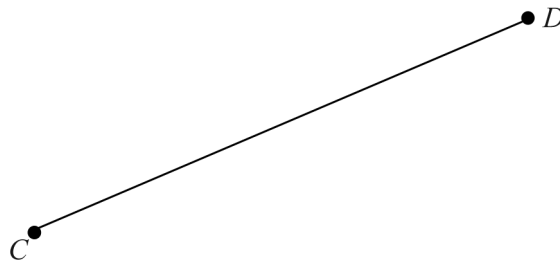
Answer (b)cm² [2]

- 3 The diagram shows a map of four villages A , B , C and D .



- (a) Using a scale of 1 cm to represent 2.5 km, make an accurate drawing of the map on the space provided.

CD has been drawn for you.



Scale: 1 cm: 2.5 km

[4]

- (b) Measure and write down the size of the angle ABC .

Answer (b) $^\circ$ [1]

- (c) Find the **actual** distance from village B to C .

Answer (c) km [2]

- (d) Describe the locus of all points that are equidistant from C and D .

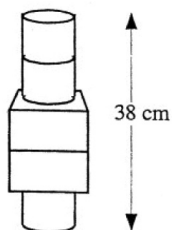
Answer (d)
 [2]

- (e) Another village E is located on the map such that it is,

- 7.5 km away from village B
- equidistant from BC and CD
- closer to A than C .

On your scale drawing, by construction, mark the location of the village E . [3]

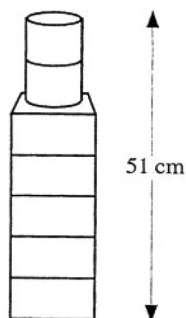
- 4 Neo builds toy towers using identical cylinders and identical cuboids.
A tower of height 38 cm is built from three cylinders and two cuboids as shown below.
Let x cm be the height of one cylinder and y cm be the height of one cuboid.



- (a) Form an equation connecting x and y .

Answer (a) [2]

- (b) Neo then builds the second tower of height 51 cm using **two** cylinders and **five** cuboids as shown.



This new information is represented by the equation

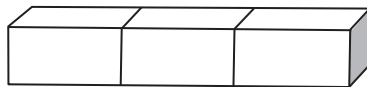
$$2x + 5y = 51.$$

Use the above equation and the one in (a)(i) to find the height of a cylinder and a cuboid.

Answer (b) Height of cylinder = cm

Height of cuboid = cm [4]

- 5 (a) Pule has 3 identical cuboids, each has length l cm and width w cm. He places the 3 cuboids together on a flat surface to make the shape shown below. The distance around the base of this shape is given by an equation $6l + 2w = 72$, while the area of the base of the shape is given by an equation $3lw = 243$.



- (i) Using equations for the distance and the area, form an equation in w and show that it reduces to $w^2 - 36w + 243 = 0$

[3]

- (ii) Solve the equation in (b)(i).

Answer (b)(ii) $w = \dots\dots\dots$ or $\dots\dots\dots$ [2]

- (iii) Find the length of the cuboid.

Answer (b)(iii) $\dots\dots\dots$ cm [2]

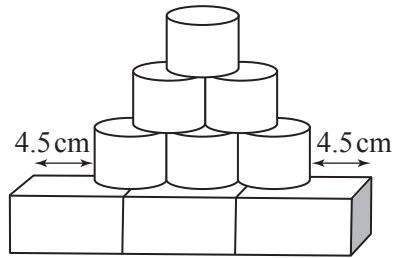
- (iv) The height of the cuboid is 4 cm.

Calculate the volume of each cuboid.

Answer (b)(iv) $\dots\dots\dots$ cm² [2]

- (b) Thabiso places 6 cylinders on the three cuboids as shown.

The first row of cylinders leaves an equal distance of 4.5 cm on either side.



Find the diameter of each cylinder.

Answer (c) cm [2]

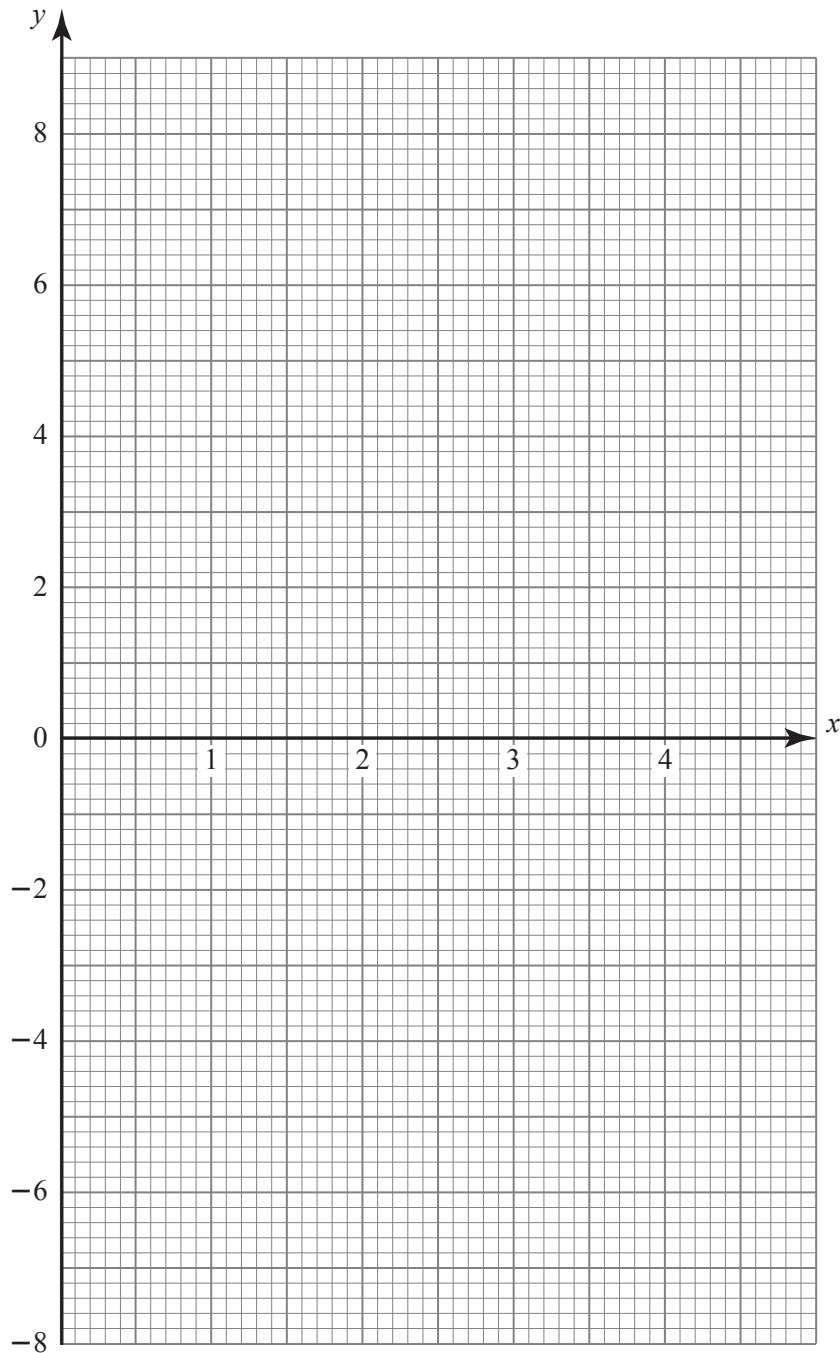
- 6 The table shows some values of x and y for $y = (3 - x)(3 + x)$.

x	0	1	1.5	2	2.5	3	3.5	4
y	9	8	6.8	5	2.8	0		-7

(a) Complete the table.

[1]

(b) On the grid provided plot the points in the table and join them with a smooth curve.



[3]

- (c) Use your graph to find the value of y when $x = 3.8$.

Answer (c) $y =$ [1]

- (d) On the same axes, draw the graph of $y = 2x$. [2]

- (e) Write the x value at which the line $y = 2x$ intersects the curve.

Answer (e) $x =$ [1]

- 7 The table shows information about the number of computers in the offices in an institution.

Number of computers per office	1	2	3	4	5	6
Number of offices	18	10	2	15	8	2

- (a) Find the

- (i) modal number of computers per office,

Answer (a)(i) [1]

- (ii) range for the number of computers per office,

Answer (a)(ii) [1]

- (iii) mean number of computers per office.

Answer (a)(iii) [3]

- (b) An office is chosen at random from the record.
Find the probability that the chosen office has

- (i) 3 computers,

Answer (b)(i) [1]

- (ii) 15 computers,

Answer (b)(ii) [1]

- (iii) at least 5 computers.
Give your answer in its simplest form.

Answer (b)(iii) [2]

- (c) Two offices are chosen at random without replacement.

Find the probability that one office has 2 computers and the other office has 5 computers.

Answer (c) [3]

- (d) x more offices are added to the record.

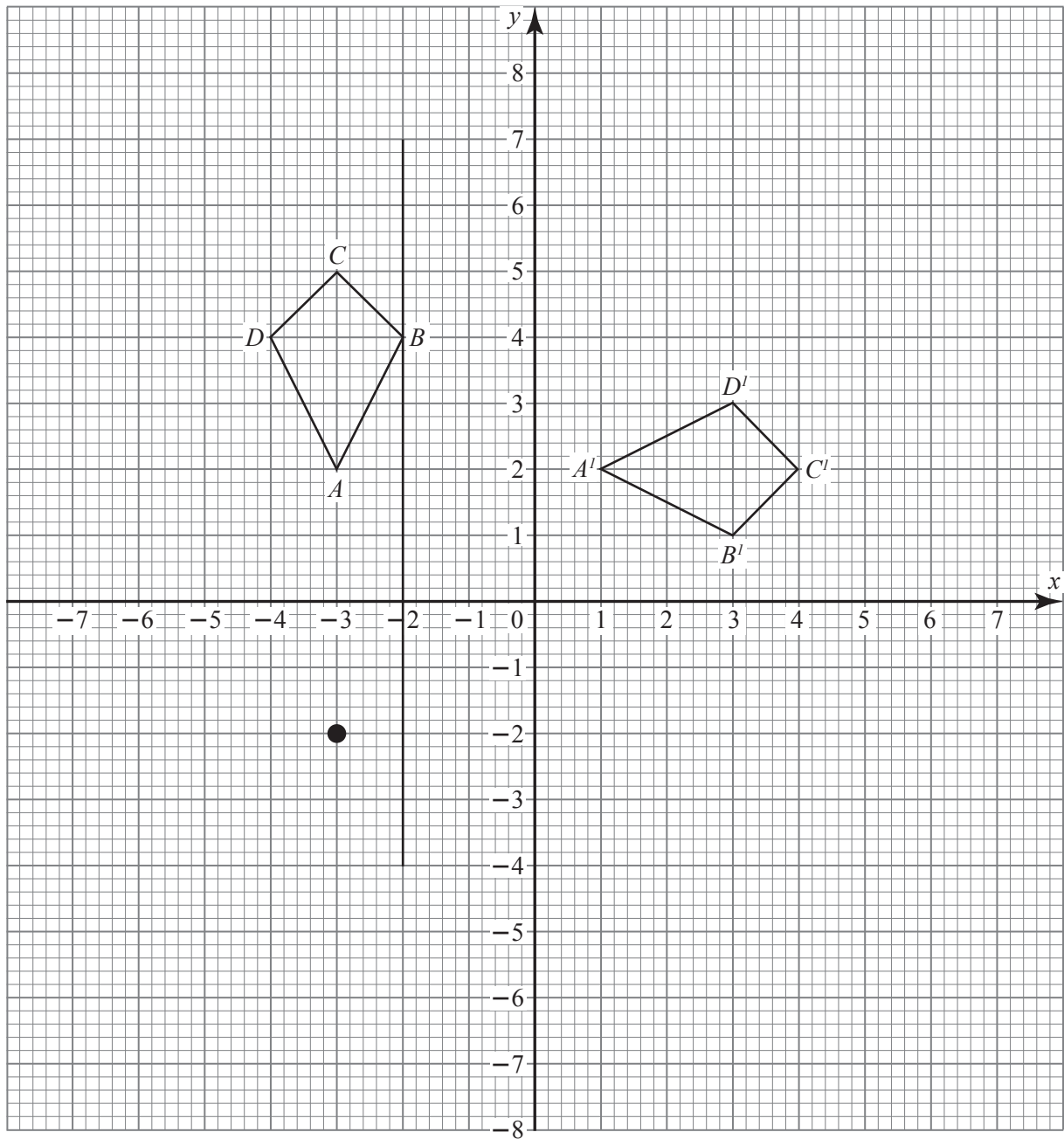
None of these x offices has 4 computers.

A pie chart is drawn from the new record.

Write, in terms of x , an expression for the sector angle representing offices with 4 computers.

Answer (d) [2]

- 8 The diagram shows a line P , a kite $ABCD$ and its image $A'B'C'D'$.



- (a) Write down the coordinates of point C .

Answer (a)(..... ,) [1]

- (b) Given that point F $(-3, -2)$ is the image of A under translation, find the translation vector.

Answer (b) $\left(\quad \quad \right)$ [2]

- (c) Draw the image of kite $ABCD$ under reflection in the mirror line P .

Label the image $A''B''C''D''$. [2]

- (d) Describe fully a **single** transformation that maps Kite $ABCD$ onto $A'B'C'D'$.

Answer (d)

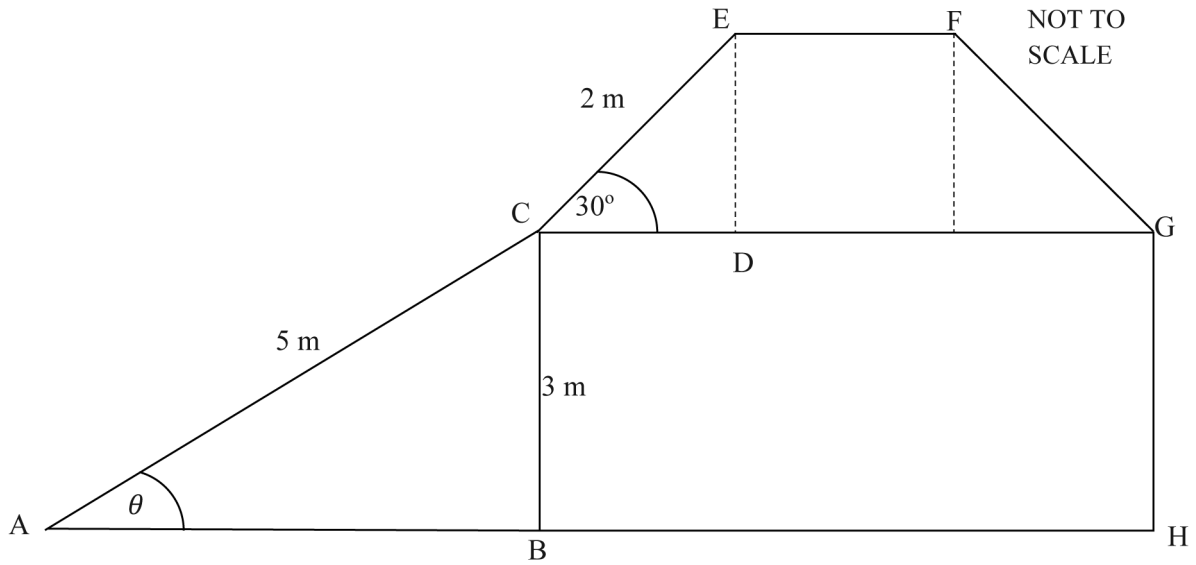
 [3]

- (e) Draw the image of kite $ABCD$ after enlargement using scale factor -2 and centre $(-3, 1)$.

Label the image $A'''B'''C'''D'''$. [2]

- 9 The diagram shows a ladder, AC , leaning against a house, $BCEFGH$.

$AC = 5\text{ m}$, $BC = 3\text{ m}$, $CE = 2\text{ m}$ and $\angle ECD = 30^\circ$.



- (a) Calculate the length

(i) AB ,

Answer (a)(i) m [3]

(ii) ED .

Answer (a)(ii) m [2]

- (b) Find the height of the house.

Answer (b) m [1]

- (c) Calculate the angle θ .

Answer (c) [2]

10 (a) A packet of beans has a mass of 250 g correct to the nearest gram.

(i) Write the least possible mass of the packet.

Answer (a)(i) g [1]

(ii) Calculate the upper bound for the mass of 8 packets.

Answer (a)(ii) g [2]

(iii) A shopkeeper buys 24 packets of beans for M156.

Calculate

(a) the cost price of each packet,

Answer (a)(iii)(a) M [1]

(b) the percentage profit if all the packets are sold for M204.

Answer (a)(iii)(b) % [3]

(b) Bank A offers 12% simple interest **per annum**.

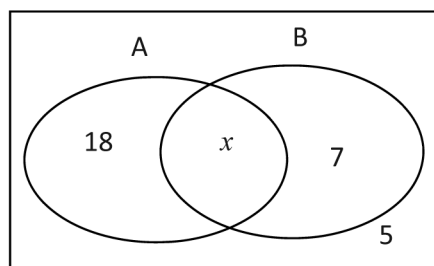
Bank B offers 1% compound interest **per month**.

If M7 000 is invested in each bank for **three** months, which bank will give greater interest?

Show all the workings clearly.

[4]

- 11 A teacher asks 40 students whether they like Apples (A) or Bananas (B).
18 students like apples only, 7 like bananas only and 5 like neither.



The Venn diagram represents the information.

- (a) Find the value of x .

Answer (a) [2]

- (b) Write the value of $n(A \cap B')$.

Answer (b) [1]

- (c) Describe the set $n(A \cap B)$ in words.

Answer (c)
..... [2]

- (d) Use set notation to represent the number of students who like one type of fruit only.

Answer (d) [2]

- (e) Write in its simplest form, the ratio of number of students who like apples only to the total number of students.

Answer (e) [2]

